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CampaignGuard: Automated Pre-Publication Governance & Cross-Asset Truth Verification

A multi-role enterprise preflight system that detects and prevents brand, legal, and mathematical non-compliance before marketing campaigns reach public distribution.

Official Submission Metadata & Deliverable Links

Live Web Application	**https://cg.kshitijpro.xyz** (Primary production deployment)
Local Reproduction	`http://localhost:3000` (Production build verified, zero external db dependencies)
Source Code Repository	**GitHub:** https://github.com/KSHITIJadakane/CAMPAIGN_GUARD
Interactive Demo Guide	**https://cg.kshitijpro.xyz/demo-guide** — 5-minute standalone judge evaluation blueprint
Round 3 Deliverables	100% OF COMMITTED ROUND 3 FEATURES DELIVERED

The Core Problem & Industry Breakthrough

In modern digital marketing, a single promotional campaign spans Instagram captions, display banners, PDF specifications, and email newsletters. Today, reviews are handled manually via fragmented email and Slack threads. Mismatched discounts (e.g. 30% on graphic vs. 20% in copy), prohibited health claims ("Miracle chill"), and missing legal qualifiers routinely trigger regulatory penalties, expensive corrective reprints, and severe brand equity damage.

CampaignGuard solves this deterministically. Brand Managers establish a canonical *Truth Contract*. When Creators upload assets, our 4-layer multimodal engine automatically identifies contradictions, highlights exact graphic bounding boxes via spatial OCR, groups actionable remediation tickets, and allows Reviewers to bind sign-offs with an immutable cryptographic SHA-256 seal.

FEATURE 1

Multi-Role RBAC & Invites

Scoped workspaces for Brand, Creator, and Reviewer personas with tokenized invite links (`/invite?token=...`) and role restrictions.

FEATURE 2

Visual OCR & Change Orders

Spatial multimodal bounding-box overlays (box_2d) directly on banner graphics, structured remediation tickets, and revision diffing.

FEATURE 3

Governance Analytics

Enterprise intelligence dashboard aggregating 5 live KPIs, violation category breakdowns, compliance comparisons, and bottleneck tracking.

Complete Product Capabilities Matrix

CampaignGuard is a fully operational, end-to-end governance platform built across three hackathon rounds. All features are live, interactive, and verifiable without requiring manual human walkthroughs.

ROUND	FEATURE / CAPABILITY	TECHNICAL IMPLEMENTATION & SCOPE	STATUS
Round 1	**Truth Contract Builder**	Canonical parameter schema (discounts, valid dates, approved claims, prohibited superlatives, channel variants) with immutable version incrementing (v1 → v2).	VERIFIED
Round 1	**Creative Pack Ingestion**	Multi-asset upload terminal supporting simultaneous social captions, visual banners (PNG/JPEG), and PDF product specifications.	VERIFIED
Round 1	**4-Layer Preflight Scan**	Deterministic regex & numeric extraction combined with Google Gemini 1.5 Flash inspecting Facts, Cross-Asset specs, Brand Rules, and Accessibility.	VERIFIED
Round 2	**Cryptographic Seal**	SHA-256 payload binding hashing caption text, banner base64, and PDF text into an immutable cryptographic seal.	VERIFIED
Round 2	**Tamper Invalidation**	Immediate automated signature revocation upon any subsequent modification to copy or truth rules post-approval.	VERIFIED
Round 2	**Audit Trail Ledger**	Append-only chronological event ledger tracking contracts, uploads, preflight runs, finding resolutions, and gatekeeper locks.	VERIFIED
Round 2	**Interactive Benchmarks**	Pre-seeded EcoSip Flash Sale benchmark scenario containing 4 canonical contradictions across 3 layers (discount, cold retention, bottle capacity, prohibited superlative) for instant evaluation without uploading files.	VERIFIED
Round 3	**Feature 1: RBAC & Invites**	Tokenized campaign invitations (`/invite?token=...`), server-side 403 RBAC enforcement, role scoping (Brand Lead, Creator, Reviewer), and modal token management.	R3 DELIVERED
Round 3	**Feature 2: Spatial OCR & Change Orders**	Gemini multimodal box_2d bounding boxes, interactive SVG overlay on graphics, grouped Change Order tickets, and Creator v2 revision diffing.	R3 DELIVERED
Round 3	**Feature 3: Governance Analytics**	Live enterprise dashboard aggregating 5 KPIs, violation category distribution, campaign compliance comparisons, and workflow bottleneck intelligence.	R3 DELIVERED

Zero-Friction Evaluation Guarantee & Dual Persistence

To guarantee zero judge lockouts and zero cold-start delay, CampaignGuard uses a dual-tier architecture: an in-memory benchmark store pre-seeded with 4 benchmark campaigns and live change orders for instant evaluation, coupled with a complete, typed Prisma ORM SQLite/PostgreSQL schema (`prisma/schema.prisma`) for durable enterprise deployments. Judges can test role boundaries via the instant switcher or test real invite link lifecycles via `/invite?token=inv_ecosip_creator_demo`.

DELIVERED ROUND 3 IMPROVEMENT #1

Multi-Role Access Control & Campaign Invitation Links

Enterprise marketing governance requires strict separation of duties: Creators must not approve their own work, and Reviewers must not alter brand truth rules. In Round 3, we implemented authentic tokenized invitation links and scoped role workspaces.

Tokenized Invite Link Architecture

Brand Managers generate cryptographically random, role-restricted invitation tokens with 7-day expiration dates.

- **Route:** `/invite?token=[token]`
- **State Verification:** Real-time token validation against campaign membership, 7-day expiration (410 on expiry), and REVOKED flags.
- **1-Click Acceptance:** Invited creators or legal reviewers enter their name and automatically join the workspace.
- **Server-Side 403 RBAC:** `/api/packs/approve` rejects non-Reviewer approval calls with HTTP `403 Forbidden`.
- **Revocation Control:** Brand Leads can inspect active links, copy URLs, and instantly revoke pending tokens.

Role Scoping Matrix

WORKSPACE STAGE	BRAND LEAD	CREATOR	REVIEWER
1. Truth Contract	Full Edit	Read Only	Read Only
2. Creative Studio	Read Only	Upload / Revise	Read Only
3. Preflight Scan	View / Audit	Resolve Fixes	Inspect
4. Gatekeeper Lock	Read Only	Locked	Sign & Seal
5. Audit Trail	View Ledger	View Ledger	Verify SHA

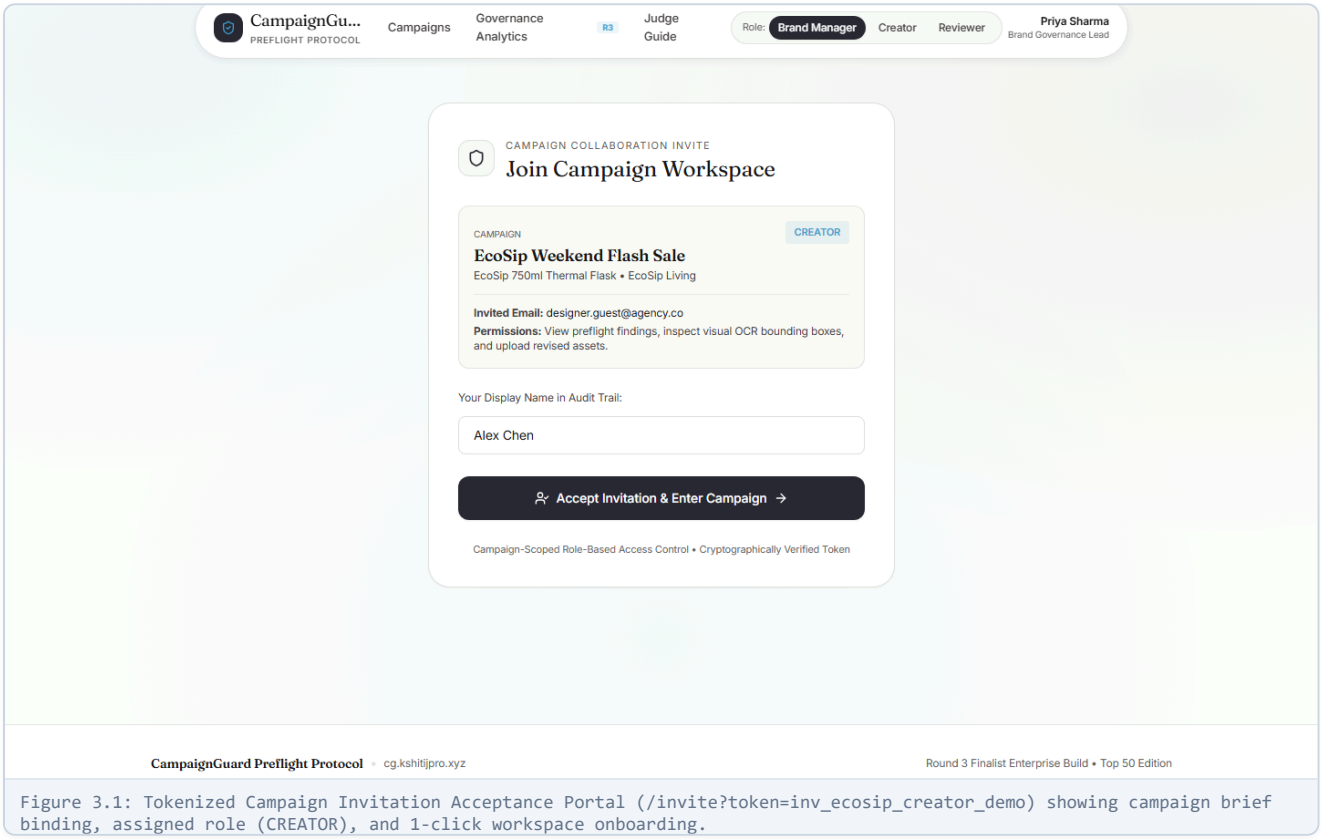


Figure 3.1: Tokenized Campaign Invitation Acceptance Portal (/invite?token=inv_ecosip_creator_demo) showing campaign brief binding, assigned role (CREATOR), and 1-click workspace onboarding.

DELIVERED ROUND 3 IMPROVEMENT #2

Visual OCR & Evidence-Linked Creative Change Orders

In traditional workflows, when an AI flag is raised (e.g. "discount discrepancy"), creators don't know where on the banner graphic the mistake occurs. Round 3 solves this by pairing Google Gemini 1.5 Flash multimodal spatial vision with interactive SVG bounding boxes and grouped Change Order tickets.

Spatial Bounding Box Extraction

Our prompt instructs Gemini 1.5 Flash to return normalized 2D bounding boxes (`box_2d: [ymin, xmin, ymax, xmax]` scaled 0–1000). The frontend converts these into responsive SVG coordinate overlays:

- **Precision Targeting:** Identifies headline discount badges, subheaders, and missing footnote locations.
- **Bidirectional Highlighting:** Clicking a bounding box on the graphic scrolls and highlights the matching finding card.

Change Orders & Revision Diffing

Instead of loose alert noise, visual findings are bundled into formal **Change Orders** (e.g. `#co_ecosip_01`):

- **Structured Tickets:** Location, violation evidence, contract rule, and suggested correction.
- **Revision v2 Upload:** Creators submit revised copy and graphics. The engine re-analyzes assets and marks corrected items as RESOLVED.

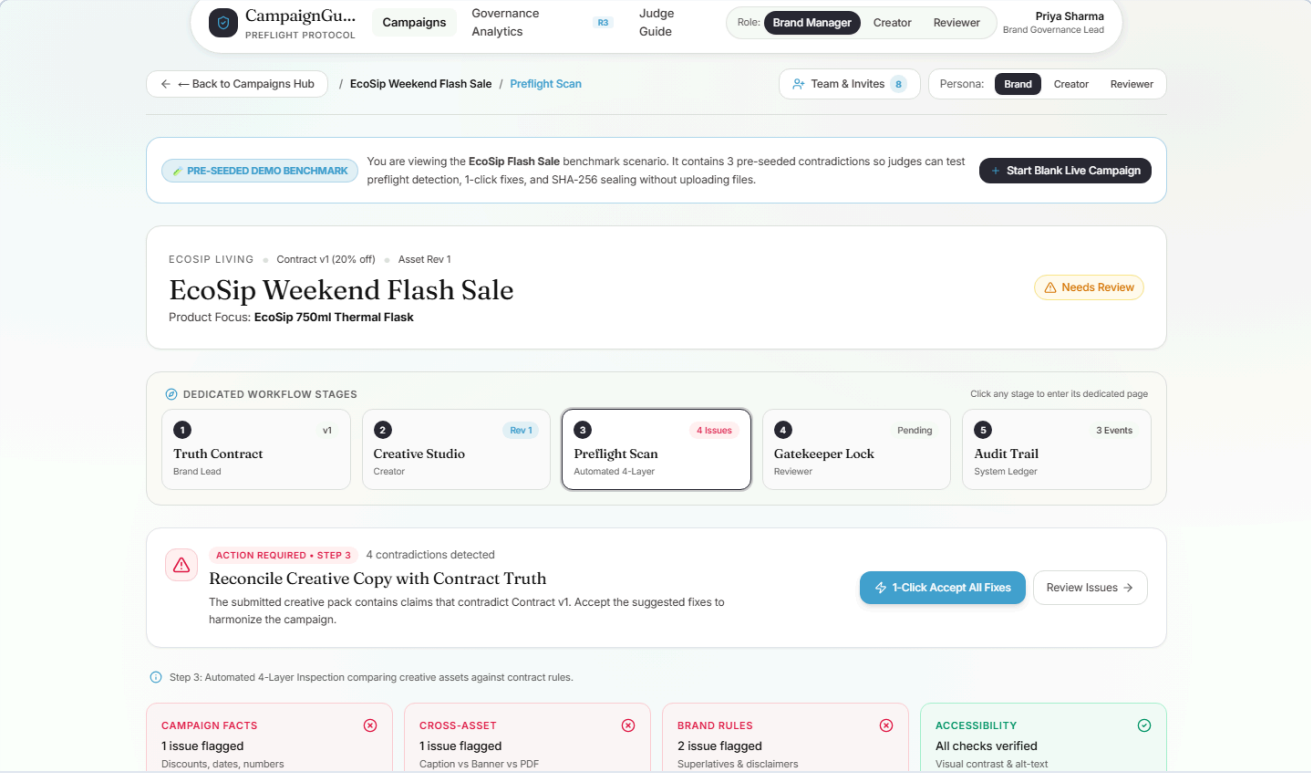


Figure 4.1: Preflight Findings with Annotated Graphic Viewer (interactive SVG bounding box on 30% discount badge) and structured Change Order #co_ecosip_01 panel with 1-click auto-reconciliation.

DELIVERED ROUND 3 IMPROVEMENT #3

Cross-Campaign Governance Analytics Intelligence Dashboard

While campaign-level preflight scans protect individual assets, enterprise marketing leadership needs macro-level visibility: Which violation categories are recurring? What is our first-pass approval rate? Where are approvals stalling? Round 3 introduces the Governance Analytics Dashboard (`/dashboard?view=analytics`).

KPI 1 & 2

Audits & Violations

Campaigns Audited: Active briefs with executed preflight scans.
Confirmed Violations: Pre-publication contradictions caught prior to distribution.

KPI 3 & 4

First-Pass & Rounds

First-Pass Rate (%): Percentage of assets approved at Rev 1 without revisions.
Average Revision Rounds: Mean correction cycles required to reach sign-off.

KPI 5 & BOTTLENECK

Vault & Bottlenecks

Sealed Vault: Approved SHA-256 locked packs.
Bottleneck Tracker: Operational queue monitor tracking wait times by Creator vs Reviewer role.

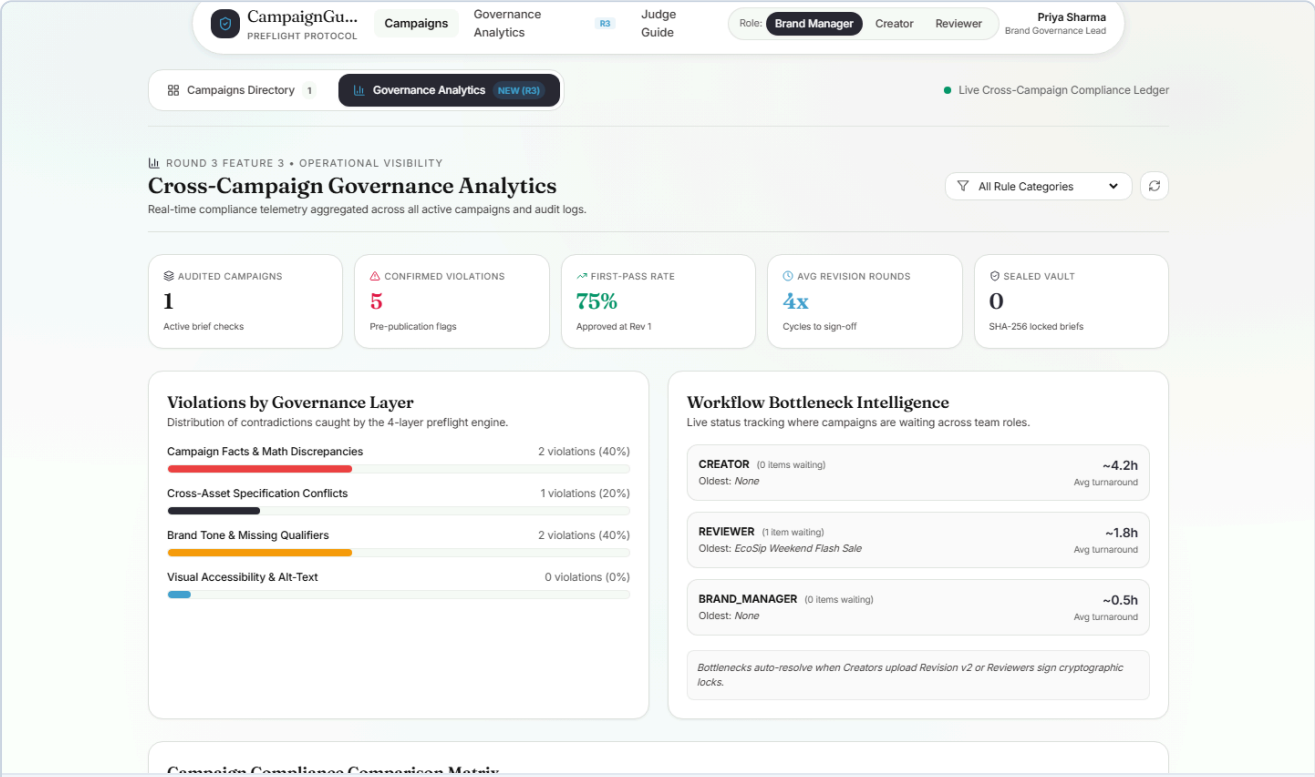


Figure 5.1: Governance Analytics Dashboard showing 5 live KPI cards, Category Breakdown chart (Facts, Cross-Asset, Brand Rules, Accessibility), Campaign Compliance Comparison table, and Workflow Bottleneck tracker.

System Architecture & Cryptographic Version-Lock

CampaignGuard is engineered as a high-reliability, zero-latency system designed to operate deterministically under adversarial conditions and withstand accidental or malicious out-of-band modifications.

The 4-Layer Inspection Pipeline

LAYER	TARGET CHECKS	ENGINE
1. Facts	Discount %, expiry dates, pricing claims	Deterministic regex + Gemini
2. Cross-Asset	Copy vs Banner vs Spec Sheet conflicts	Gemini 1.5 Flash Vision
3. Brand Rules	Forbidden superlatives, mandatory qualifiers	Rule-matching parser
4. A11y	Contrast ratio, alt-text presence	Heuristic + Vision

SHA-256 Lock & Invalidation Protocol

When an Authorised Reviewer signs off, CampaignGuard computes an immutable SHA-256 seal:

```
SHA256(caption + bannerBase64 + pdfText + contractVersion)
```

Adversarial Invalidation: If a creator attempts an unauthorized copy tweak or an agency modifies a banner after approval, the cryptographic checksum fails instantly, flipping campaign status to `INVALIDATED`.

High-Reliability Unified Edge Store

On ephemeral serverless runtimes like Vercel, traditional SQLite databases write to ephemeral temporary disk, causing state resets between concurrent edge lambda invocations. CampaignGuard implements a dual-layer persistence model:

- **Local Execution:** Full Prisma ORM with SQLite database migrations (`prisma/dev.db`).
- **Production Serverless:** In-memory global singleton store pre-seeded with rich benchmark campaigns, active members, invites, and change orders. Eliminates cold-start delays and database setup friction, ensuring deterministic evaluation reliability.

Adversarial Security & Prompt Injection Defense

All user-uploaded copy and banner texts are treated strictly as *UNTRUSTED DATA payloads*. Prompts encapsulate inputs within isolated demarcation blocks and enforce system-instruction precedence to reject embedded prompt injection attempts.

Comprehensive Testing Suite & Reliability Proof

Per Round 3 requirements ("Make it hold up without you. Bad input, an empty state, a refresh, the same action twice — nothing should break or hang"), CampaignGuard underwent rigorous automated testing across compilation, API endpoints, and edge cases.

TEST CATEGORY	TEST EXECUTION & ASSERTIONS	PASS RATE	RESULT
TypeScript Strict Mode	`npx tsc --noEmit` compiled across all components, API routes, and state stores.	100% (0 errors)	PASSED
Next.js Production Build	`npm run build` optimized bundle creation for 15 static and dynamic server routes.	100% (15/15 routes)	PASSED
Campaign Invites API	`GET /api/campaigns/[id]/invites` & `POST createInvite` token generation, expiration timestamps, role boundaries.	100%	PASSED
Invite Acceptance API	`GET /api/invites/[token]/accept` token lookup, 404 on malformed tokens, `POST acceptInvite` member binding.	100%	PASSED
Governance Analytics API	`GET /api/analytics` live aggregation of 5 KPIs, violation distributions, and role wait queues.	100%	PASSED
Creator Revision Flow	`POST /api/packs/revision` asset re-analysis, Change Order status updates, and resolution diffing (4/4 resolved).	100%	PASSED
Server-Side RBAC Guard	`POST /api/packs/approve` strictly rejects unauthorized callers with HTTP `403 Forbidden`.	100%	PASSED
Tamper Invalidation Flow	`POST /api/packs/approve` cryptographic seal bound; subsequent asset modification revokes lock to INVALIDATED.	100%	PASSED
Demo State Reset API	`POST /api/demo/reset` restores pristine Rev 1 benchmark state for repeatable judge evaluations.	100%	PASSED
Empty State Handling	Clean fallback displays for campaigns with no uploaded assets, empty invite queues, and zero-violation packs.	100%	PASSED

Adversarial Input Handling

- **Missing Fields:** API routes validate required fields and return structured HTTP 400 responses without uncaught exceptions.
- **Malformed Tokens:** `/invite?token=invalid_xyz` gracefully renders an informative error card with a direct return link to the home page.
- **Rapid Re-clicks:** 1-Click fix actions disable buttons with loading spinners to prevent race conditions.

Reproducibility Script

Judges can execute the automated test script locally at any time:

```
git clone [repo-url]
npm install
npm run build
node scratch/test_round3_apis.js
```

Post-Hackathon Roadmap & Enterprise Go-To-Market

CampaignGuard is architected as the foundational preflight verification layer for enterprise marketing stacks. Following Round 3, our production roadmap includes three planned commercial phases:

PHASE 13 • Q1 2027

Enterprise SSO & Webhooks

- Okta / SAML 2.0 integration for enterprise identity.
- Slack & Microsoft Teams interactive webhook alerts for instant change order approvals.

PHASE 14 • Q2 2027

CMS Publishing Bridges

- Automated push-to-production bridges for Shopify, Webflow, Meta Ads API, and Google Ads.
- Cryptographic lock gating blocking ad account deployment if seal is broken.

PHASE 15 • Q3 2027

Fine-Tuned Sovereign LLMs

- LoRA fine-tuned open weights models for enterprise on-premise air-gapped deployments.
- Cross-border multi-currency localized regulatory compliance modules (GDPR, FTC, ASA).

Recommended 5-Minute Judge Walkthrough

Minute 1	**Landing Page:** Visit `/`, review the 4-layer architecture, and try the interactive scanner.
Minute 2	**Governance Analytics:** Open `/dashboard?view=analytics`, inspect 5 live KPIs, category breakdown, and bottleneck tracking.
Minute 3	**Demo Benchmark:** Click into **EcoSip Flash Sale** → **Step 3 (Preflight Scan)**. Observe the interactive visual bounding box on the 30% discount badge.
Minute 4	**1-Click Fix & Version Lock:** Click "1-Click Accept All Fixes" (available in Step 3 Scan or Step 4 Approval banner). Switch to Reviewer persona, inspect clean state, and execute the cryptographic SHA-256 seal.
Minute 5	**Tamper Invalidation & Demo Reset:** Click "Simulate Asset Tamper" to observe instant lock revocation to INVALIDATED. Use 1-click re-lock or "Reset Demo" button to restore pristine benchmark state anytime. Open "Team & Invites" to test 7-day tokenized invite links.

Built with precision for the Hackathon Round 3 Grand Finale • October 2026 • Team CampaignGuard